

Vel Tech Group of Institutions

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2028_NeoColab_Design and Analysis of AlgorithmsS1L3

VTU_DAA_Week 4_MCQ

Attempt : 1

Total Mark : 15

Marks Obtained : 14

Section 1 : MCQ

1. Which of the following methods can be used to solve the Knapsack problem?

Answer

All the mentioned options

Status : Correct

Marks : 1/1

2. 0/1 knapsack is based on _____method.

Answer

Dynamic programming

Status : Correct

Marks : 1/1

3. What is the primary advantage of using the Branch and Bound approach for the Knapsack problem?

Answer

It systematically explores possible solutions while pruning non-promising ones.

Status : Correct

Marks : 1/1

4. Which data structure is commonly used to implement the Branch and Bound approach for solving the Knapsack problem?

Answer

Priority Queue

Status : Correct

Marks : 1/1

5. In the Branch and Bound approach, what does the term “bounding function” refer to?

Answer

A function that estimates the upper bound on the maximum value that can be achieved from a node.

Status : Correct

Marks : 1/1

6. Which strategy is commonly used in the Branch and Bound method to decide which node to explore next?

Answer

Best-first search

Status : Correct

Marks : 1/1

7. Which of the following methods is used to prune the search space in the Branch and Bound method for the Knapsack problem?

Answer

Ignoring nodes with a weight exceeding the capacity.

Status : Correct

Marks : 1/1

8. The travelling salesman problem can be solved using _____.

Answer

Bellman - Ford algorithm

Status : Wrong

Marks : 0/1

9. Travelling salesman problem is an example of _____.

Answer

None of the mentioned options

Status : Correct

Marks : 1/1

10. How does the practical Travelling salesman problem differ from the classical Travelling salesman problem?

Answer

In the practical travelling salesman problem, each vertex can be visited more than once.

Status : Correct

Marks : 1/1

11. The traveling salesman problem involves n cities with paths connecting the cities. The time taken for traversing through all the cities without knowing in advance the length of a minimum tour is

Answer

$O(n!)$

Status : Correct

Marks : 1/1

12. In which search problem, to find the shortest path, each city must be

visited only once?

Answer

Travelling Salesman problem

Status : Correct

Marks : 1/1

13. What is the main goal of using Branch and Bound in TSP?

Answer

To minimize search space by pruning unpromising paths

Status : Correct

Marks : 1/1

14. Which value helps in deciding whether a node should be pruned in Branch and Bound?

Answer

Bound (lower bound of cost)

Status : Correct

Marks : 1/1

15. Given items with weights [10, 20, 30] and values [60, 100, 120], what is the value-to-weight ratio for the second item?

Answer

5

Status : Correct

Marks : 1/1